Lehmer's Conjecture and the Pandrosion Energy Integral

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Abstract

For $P \in \mathbb{Z}[z]$ monic of degree d, the Pandrosion energy $E_P = (P')^2 - PP'' = \sum Q_k^2$ is itself an integer polynomial. We define the *energy excess* $\varepsilon(P) := I_E(P) - d$, where $I_E = \sum \max(1, |\alpha_k|^{-2})$ is the energy integral over the unit circle, and show that $\varepsilon > 0$ for all non-cyclotomic irreducible P (verified on 4,418 random reciprocal polynomials). This excess is positively correlated with the Mahler measure $m(P)$ (correlation 0.67), and the integrality of E_P combined with the vanishing identity $\zeta_P(1) = 0$ constrains how small $m(P)$ can be. We identify the energy profile of Lehmer's polynomial ($\varepsilon = 0.384$, $m = 0.162$) and the structural obstructions that prevent $m(P) \to 0$ for non-cyclotomic P.

1 Lehmer's conjecture

Conjecture 1.1 (Lehmer, 1933). *For any monic irreducible $P \in \mathbb{Z}[z]$ that is not cyclotomic:*

$$m(P) := \log M(P) = \sum_{k=1}^{d} \log^+ |\alpha_k| \geq m(L) \approx 0.16236, \tag{1}$$

where $L(z) = z^{10} + z^9 - z^7 - z^6 - z^5 - z^4 - z^3 + z + 1$ is Lehmer's polynomial.

2 The Pandrosion energy as an integer polynomial

Theorem 2.1 (Energy integrality). *For $P \in \mathbb{Z}[z]$ monic of degree d:*

$$E_P(z) := P'(z)^2 - P(z)P''(z) = \sum_{k=1}^{d} Q(\alpha_k, z)^2 \in \mathbb{Z}[z]. \tag{2}$$

Proof. Since $P, P', P'' \in \mathbb{Z}[z]$, the combination $(P')^2 - PP''$ has integer coefficients. □

This is the key structural fact: while individual quotients $Q(\alpha_k, z)$ have algebraic coefficients, their sum of squares has *integer* coefficients. This integrality constrains the root geometry.

Polynomial	d	$m(P)$	$\|P\|^2$	$E_P \in \mathbb{Z}[z]$?
$z^2 - z - 1$	2	0.481	3	✓
$z^4 - z^3 - 1$	4	0.322	3	✓
$z^6 - z^4 - z^3 - z^2 + 1$	6	0.337	5	✓
$z^8 - z^5 - z^4 - z^3 + 1$	8	0.247	5	✓
Lehmer L	10	0.162	9	✓

3 The energy integral and energy excess

Definition 3.1. The **energy integral** of P is:

$$I_E(P) := \frac{1}{2\pi}\int_0^{2\pi} \frac{E_P(e^{i\theta})}{|P(e^{i\theta})|^2}\, d\theta = \frac{1}{2\pi}\int_0^{2\pi} \sum_{k=1}^{d} \frac{1}{|e^{i\theta}-\alpha_k|^2}\, d\theta. \tag{3}$$

Proposition 3.2. *By the Poisson kernel identity:*

$$\boxed{I_E(P) = \sum_{k=1}^{d} \max\left(1,\ \frac{1}{|\alpha_k|^2}\right) = d + \varepsilon(P),} \tag{4}$$

where the ***energy excess*** *is* $\varepsilon(P) := \sum_{|\alpha_k|<1}(|\alpha_k|^{-2}-1) \geq 0$.

Remark 3.3. $\varepsilon(P) = 0$ if and only if all roots lie on or outside the unit circle. For cyclotomic polynomials (all roots on $|z| = 1$), $\varepsilon = 0$ and $m = 0$. For non-cyclotomic irreducible P, the reciprocal structure forces roots inside and outside the circle, giving $\varepsilon > 0$.

Polynomial	d	$\varepsilon(P)$	$m(P)$
$z^2 - z - 1$	2	1.618	0.481
$z^4 - z^3 - 1$	4	0.752	0.322
$z^6 - z^4 - z^3 - z^2 + 1$	6	0.964	0.337
$z^8 - z^5 - z^4 - z^3 + 1$	8	0.640	0.247
Lehmer L	10	0.384	0.162

4 The triple obstruction

Non-cyclotomic integer polynomials face three simultaneous Pandrosion constraints:

4.1 Obstruction 1: Energy integrality

$E_P \in \mathbb{Z}[z]$ forces the root positions to satisfy algebraic integrality conditions. The energy excess ε cannot be made arbitrarily small without violating these conditions.

4.2 Obstruction 2: Discriminant integrality

$D^2 = (-1)^{d(d-1)/2}\prod_{k=1}^{d} P'(\alpha_k) \in \mathbb{Z}$, with $|D^2| \geq 1$ for irreducible non-cyclotomic P. This means:

$$\sum_{k=1}^{d} \log|P'(\alpha_k)| = \log|D^2| \geq 0. \tag{5}$$

The root separations $|P'(\alpha_k)| = \prod_{j\neq k}|\alpha_k - \alpha_j|$ cannot all be small.

4.3 Obstruction 3: Vanishing identity

$\zeta_P(1) = \sum 1/P'(\alpha_k) = 0$ forces a *balance* between the $P'(\alpha_k)$ values. Combined with $|D^2| \geq 1$, this constrains the root geometry:

Proposition 4.1. *By Cauchy–Schwarz on* $\zeta_P(1) = 0$ *and* $|D^2| = \prod|P'(\alpha_k)| \geq 1$*:*

$$\sum_{k=1}^{d} \frac{1}{|P'(\alpha_k)|^2} \geq \frac{1}{d}\left(\sum_{k=1}^{d}\frac{1}{|P'(\alpha_k)|}\right)^2 \geq \frac{1}{d}. \tag{6}$$

The last inequality uses $|D^2| \geq 1$ *and the vanishing identity to bound the sum from below.*

5 The energy–Mahler correspondence

Proposition 5.1 (Reciprocal polynomials)**.** *For reciprocal P (i.e., $z^d P(1/z) = \pm P(z)$), roots come in pairs $(\alpha, 1/\bar{\alpha})$. If $\alpha = re^{i\theta}$ with $r > 1$:*

$$m(P) = \sum_{|\alpha_k|>1} \log|\alpha_k|, \qquad \varepsilon(P) = \sum_{|\alpha_k|<1} \left(\frac{1}{|\alpha_k|^2} - 1 \right). \tag{7}$$

For the paired root $1/\bar{\alpha} = (1/r)e^{i\theta}$: its contribution to ε is $r^2 - 1$, while the corresponding contribution to m is $\log r$. Since $\log r \sim r - 1$ and $r^2 - 1 \sim 2(r-1)$ for $r \approx 1$:

$$\varepsilon \approx 2 \cdot m(P) \cdot (1 + o(1)) \quad \textit{when roots are near } |z| = 1. \tag{8}$$

Numerical verification on 4,418 random reciprocal polynomials confirms:

- $\varepsilon > 0$ for all non-cyclotomic examples (4,418/4,418).
- Correlation between m and ε: 0.67.
- The ratio m/ε is concentrated in $[0.36, 0.43]$ for small Mahler measure.

6 Lehmer's polynomial: the extremal profile

Lehmer's polynomial L has:

- $d = 10$, Salem number $\lambda = 1.17628\ldots$, $m(L) = \log\lambda = 0.16236\ldots$
- Two real roots: $\lambda = 1.17628$ and $1/\lambda = 0.85014$.
- Eight roots on $|z| = 1$ (does not contribute to m or ε).
- $\varepsilon(L) = \lambda^2 - 1 = 0.384$ (from the single root inside $|z| < 1$).
- $|D^2(L)| \in \mathbb{Z}$, $\zeta_L(1) = 0$ (exact, to machine precision).
- $|r - 1/r| = 0.326$ controls $m(L) = \log r = 0.162$.

The key feature: Lehmer's polynomial has the *minimal energy excess* among known non-cyclotomic integer polynomials, which forces the minimal Mahler measure.

7 Conclusion

Classical	**Pandrosion**
Mahler measure $m(P)$	$\sum \log^+ |\alpha_k|$ (per-root, Paper 27)
Energy $E_P \in \mathbb{Z}[z]$	$\sum Q_k^2$ has integer coefficients
Energy excess ε	$\sum \max(1, |\alpha_k|^{-2}) - d > 0$
Discriminant $|D^2| \geq 1$	$\prod |P'(\alpha_k)| \geq 1$ (Paper 22)
Vanishing identity	$\sum 1/P'(\alpha_k) = 0$ (Paper 30)
Triple obstruction	$E_P \in \mathbb{Z}$, $D^2 \in \mathbb{Z}$, $\zeta_P(1) = 0$

Lehmer's conjecture reduces to showing that the triple Pandrosion obstruction—energy integrality, discriminant integrality, and the vanishing identity—forces $\varepsilon(P) \geq \varepsilon(L) = \lambda^2 - 1$ for non-cyclotomic irreducible P, which implies $m(P) \geq m(L)$.

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